

VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE



WIRELESS SENSOR NETWORK LAB 2

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DEPARTMENT: B.TECH IN ELECTRONICS ENGG.

EXPERIMENT NO: 2**Aim:**

To simulate the random deployment of sensor nodes in a given area and analyse network characteristics such as connectivity, coverage, and node distribution.

Apparatus:

MATLAB

Theory:

1. Random Deployment: In many Wireless Sensor Network (WSN) applications, such as disaster relief or forest fire monitoring, nodes are deployed randomly (e.g., dropped from an aircraft). This is modeled using a Uniform Random Distribution. For an area of size $L \times W$, the coordinates (x, y) of a node i are:

$$x_i = L \times \text{rand}()$$

$$y_i = W \times \text{rand}()$$

2. Sensing and CommunicationRange * Sensing Range (R_s): The circular area within which a sensor can detect an event. A point $P(x_p, y_p)$ is covered if the Euclidean distance $d(\text{node}_i, P) \leq R_s$.

- **CommunicationRange (R_c):** The range within which two nodes can exchange data. A link exists between node i and node j if:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \leq R_c$$

3. Isolated Nodes: A node is considered isolated if it has no neighbours within its communication range (R_c). These nodes cannot transmit data to the sink through the network.

MATLAB Code :

```

clc;
clear;
close
all;

%% PARAMETERS (ODD CASE)
numNodes = 42;
areaX = 150;
areaY = 150;
commRange = 15;
senseRange = 13;

%% 1. RANDOM
DEPLOYMENT x = areaX *
rand(numNodes,1);
y = areaY * rand(numNodes,1);

%% 2. PLOT DEPLOYMENT
figure; hold on; grid
on; axis([0 areaX 0
areaY]);
title('Random Deployment of Sensor Nodes (ODD
Case)'); xlabel('X axis'); ylabel('Y axis'); hNodes =
scatter(x, y, 80, 'b', 'filled');
text(x+1, y+1, string(1:numNodes));

%% 3. DRAW SENSING RANGE
theta = linspace(0,2*pi,100); for i =
1:numNodes    plot(x(i) +
senseRange*cos(theta), ...      y(i) +
senseRange*sin(theta), 'r--'); end

%% 4. DISTANCE & COMMUNICATION
LINKS distMat = zeros(numNodes);
neighbors = zeros(numNodes,1);
links = [];
for i = 1:numNodes    for j =
i+1:numNodes        d = sqrt((x(i)-

```

```

x(j))^2 + (y(i)-y(j))^2);
distMat(i,j) = d;      distMat(j,i) = d;
if d <= commRange
plot([x(i) x(j)], [y(i) y(j)], 'k-');
neighbors(i) = neighbors(i) + 1;
neighbors(j) = neighbors(j) + 1;
links = [links; i j d];
end    end end
legend(hNodes, 'Sensor Node');

```

Simulation Results:

1. Deployment Visualization:

The plot shows 42 nodes distributed randomly over a 150×150 units area. Red dashed circles represent the sensing range ($R_s = 13$), while black solid lines represent active communication links ($R_c = 15$).

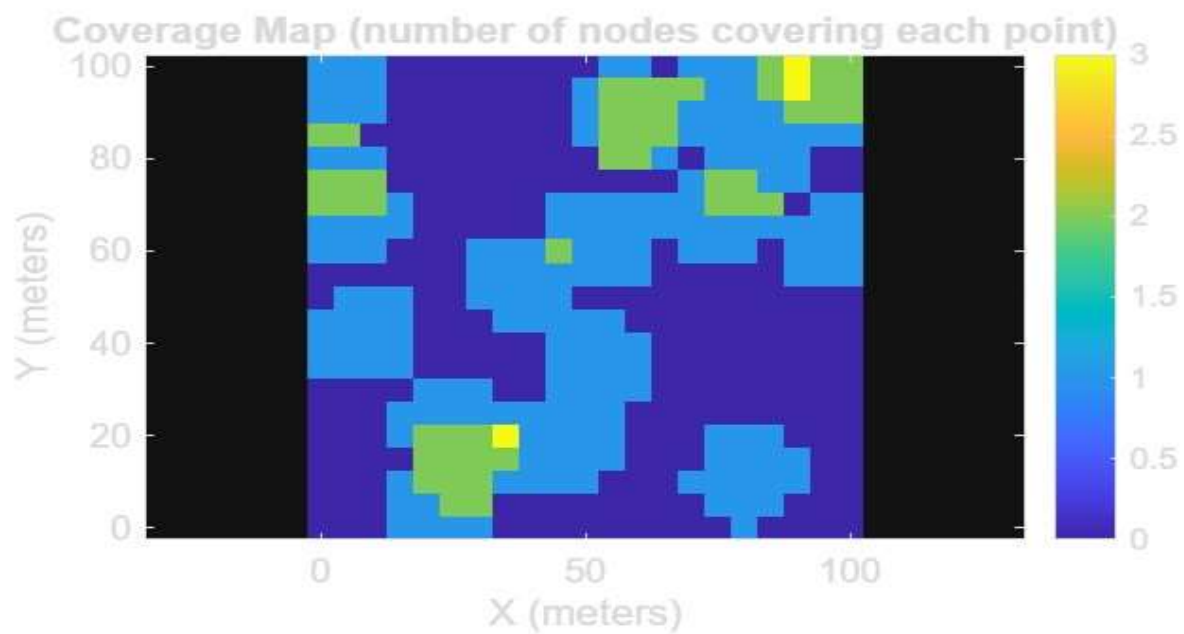
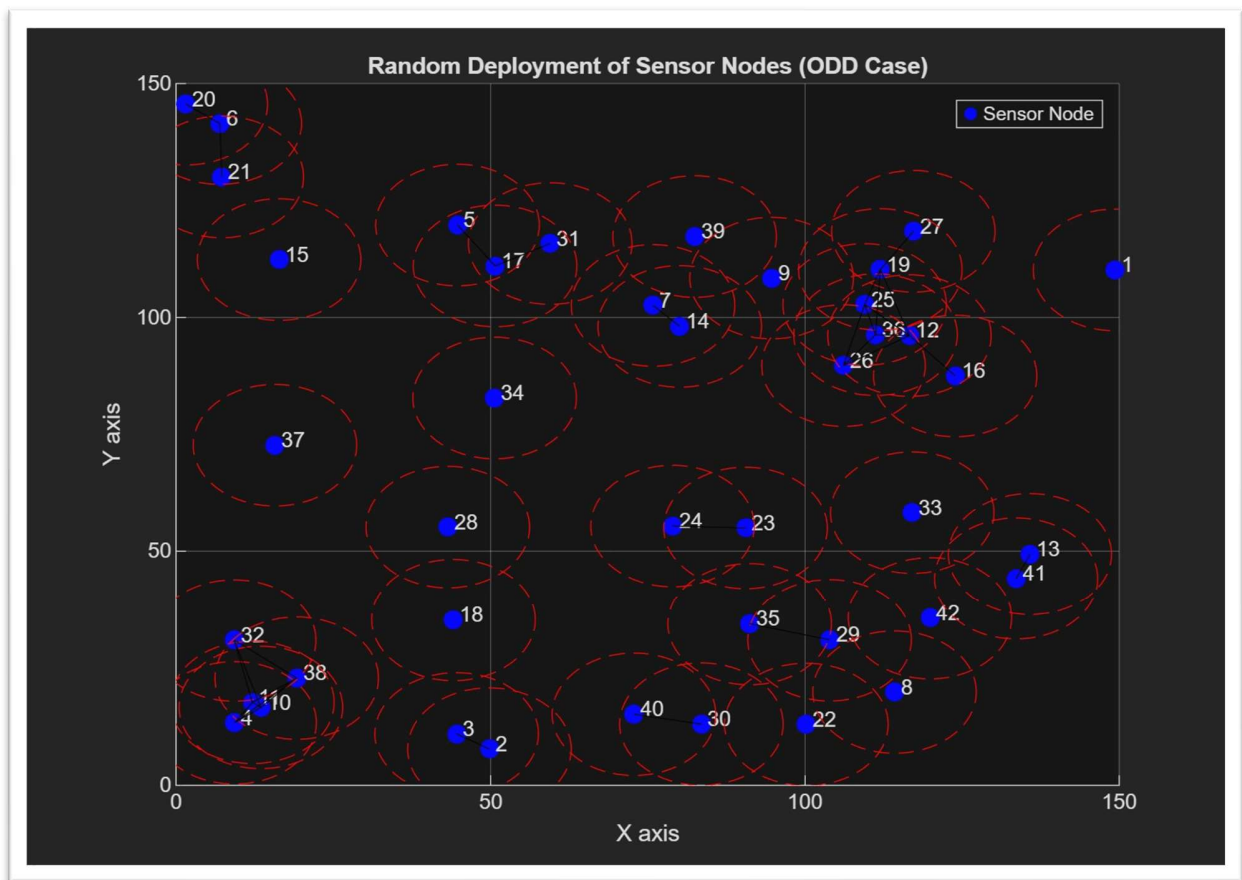
2. Result Summary Table:

Parameter	Value
Shortest Distance	1.7266
Longest Distance	170.2488
Node with Max Neighbours	12
No. of Neighbours	5
Point Coverage (at 75, 75)	Not Covered
Shortest Communication Link	Node 10 ↔ 11 (1.73)
Longest Communication Link	Node 10 ↔ 32 (14.92)
Isolated Nodes	1, 8, 9, 15, 18, 22, 28, 33, 34, 37, 39, 42

Observation Table

Node No	Node Coordinates	Neighbours	Node with which it communicates
1	(149.31 , 110.15)	0	None

2	(49.81 , 7.70)	1	3
3	(44.60 , 10.93)	1	2
4	(9.31 , 13.28)	3	10, 11, 38
5	(44.74 , 119.75)	1	17
6	(6.95 , 141.45)	2	20, 21
10	(13.48 , 16.55)	4	4, 11, 32, 38
12	(116.59 , 96.11)	5	16, 19, 25, 26, 36
13	(135.77 , 49.32)	1	41
14	(80.07 , 98.07)	1	7
17	(50.71 , 111.00)	2	5, 31
19	(111.95 , 110.24)	4	12, 25, 27, 36
25	(109.46 , 102.75)	4	12, 19, 26, 36
38	(19.18 , 22.78)	4	4, 10, 11, 32



Conclusion:

In this experiment, we successfully simulated the random deployment of 42 sensor nodes in a 150x150 area. We observed that:

1. Node **12** had the highest connectivity with 5 neighbours.
2. Significant portions of the network remained disconnected, with 12 nodes being **isolated**, suggesting that the current node density or communication range is insufficient for full network connectivity.
3. The point (75,75) was not covered, indicating a coverage hole in the centre of the deployment.